

DESIGN PROPOSAL (DRAFT)

Silicon Photonic Interferometers: From Design to Data Analysis

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Abstract: This document proposes eight unbalanced interferometers and one calibration structure on the 220 nm silicon-on-insulator (SOI) platform, with the goal of extracting the group index n_g of a 500 nm $\times$ 220 nm strip waveguide for TE polarization. The set includes five Mach-Zehnder interferometers (MZIs) with Y-branch splitters and path-length differences ΔL from 50 to 250 μm , one replica to assess fabrication variability, and two Michelson interferometers with a broadband directional coupler (BDC) and loop mirrors. Waveguide simulations in Lumerical MODE give $n_{\text{eff}} = 2.445$ and $n_g = 4.182$ at 1550 nm. Circuit simulations in Lumerical INTERCONNECT confirm free spectral ranges (FSR) between 2.3 and 11.5 nm, well within the measurement bandwidth (≈ 44 nm at -3 dB), and recover the n_g of the waveguide model for the MZIs. In the Michelson interferometers, the BDC phase adds an extra term that is removed by combining two devices with different ΔL . The expression for n_g as a function of the FSR is derived, and the procedure to apply it to the measured data is described.

1. Introduction

Interferometers are a basic building block of many photonic integrated circuits: modulators, switches, wavelength-division multiplexing (WDM) filters and sensors. In all of them, the spectral response depends on the group index n_g of the waveguide, which sets the FSR of a filter or the delay of a delay line. Since the fabricated width and thickness differ slightly from their nominal values, it is useful to measure n_g on the chip itself and use it to calibrate the compact models of the process [1], [2].

This project takes a deliberately simple approach: it uses the default waveguide of the course and well-characterized components from the SiEPIC EBeam library, and focuses on a sound experimental design. The design objectives are:

- to design unbalanced interferometers whose FSR is smaller than the measurement bandwidth (≈ 50 nm), so that several fringes are observed;
- to extract n_g of the 500 $\times$ 220 nm TE strip waveguide and compare it with simulation;
- to study the trend of the FSR versus ΔL using five different values;
- to compare an MZI with a Michelson interferometer, in which light travels each arm twice;
- to estimate fabrication variability with a repeated device.

2. Theory

2.1. Effective index and group index

The phase accumulated by the guided mode along a waveguide of length L is βL , with the propagation constant

$$\beta(\lambda) = \frac{2\pi n_{eff}(\lambda)}{\lambda} \quad (1)$$

The group index describes the speed at which the signal envelope travels and includes both waveguide and material dispersion:

$$n_g(\lambda) = n_{eff}(\lambda) - \lambda \frac{dn_{eff}}{d\lambda} \quad (2)$$

2.2. MZI transfer function

In the MZI of Fig. 3(a), a Y-branch splits the power into two arms of lengths L_1 and $L_2 = L_1 + \Delta L$, and a second Y-branch recombines them. With ideal 50/50 splitters and propagation loss α (power), the output field is

$$\frac{E_o}{E_i} = \frac{1}{2} [e^{-j\beta L_1 - \alpha L_1/2} + e^{-j\beta L_2 - \alpha L_2/2}] \quad (3)$$

and, if the losses are equal in both arms, the transmitted intensity is

$$I_o = \frac{I_i}{2} [1 + \cos(\beta \Delta L)] \quad (4)$$

2.3. Michelson transfer function

In the Michelson interferometer of Fig. 3(b), a 2×2 broadband directional coupler (BDC) [3] splits the light into two arms terminated by loop mirrors: a Y-branch whose two outputs are joined by a waveguide, so that the light returns along the same path. Light travels each arm twice and recombines in the same BDC; the signal is collected at the second input port. With an ideal 50/50 BDC and ideal mirrors:

$$I_o = \frac{I_i}{2} [1 + \cos(2\beta \Delta L)] \quad (5)$$

2.4. Derivation of n_g from the FSR

The phase difference between the MZI arms is $\varphi(\lambda) = 2\pi n_{eff} \Delta L / \lambda$. Differentiating with respect to λ and using (2):

$$\frac{d\varphi}{d\lambda} = -\frac{2\pi \Delta L}{\lambda^2} \left(n_{eff} - \lambda \frac{dn_{eff}}{d\lambda} \right) = -\frac{2\pi n_g \Delta L}{\lambda^2} \quad (6)$$

Two consecutive minima are separated by a phase change of 2π . If the FSR is small compared with λ , $|d\varphi/d\lambda| \cdot \text{FSR} \approx 2\pi$, which gives

$$\text{FSR} \approx \frac{\lambda^2}{n_g \Delta L} \Rightarrow n_g = \frac{\lambda^2}{\Delta L \cdot \text{FSR}} \quad (7)$$

For the Michelson interferometer the phase difference is $2\beta \Delta L$, so the FSR is halved for the same ΔL :

$$n_g = \frac{\lambda^2}{2\Delta L \cdot \text{FSR}} \quad (8)$$

In (7) and (8), λ is the central wavelength between the two minima that define the FSR; this yields $n_g(\lambda)$ across the whole measured band.

2.5. Effect of the splitter in the Michelson interferometer

The BDC is an asymmetric coupler [3]: its two waveguides have different widths, so the through-port phase is not the same in both. In the Michelson interferometer each arm uses the BDC in a different order (through-cross or cross-through), which adds a phase difference $\Delta\phi_{\text{BDC}}(\lambda)$. The total phase is $4\pi n_{\text{eff}}\Delta L/\lambda + \Delta\phi_{\text{BDC}}$ and, repeating the previous derivation,

$$\frac{\lambda^2}{\text{FSR}} = 2n_g\Delta L + \delta, \quad \delta = -\frac{\lambda^2}{2\pi} \frac{d\Delta\phi_{\text{BDC}}}{d\lambda} \quad (9)$$

where δ acts as an equivalent path difference that does not depend on ΔL . With two Michelson interferometers of different ΔL , δ cancels out:

$$n_g = \frac{(\lambda^2/\text{FSR})_2 - (\lambda^2/\text{FSR})_1}{2(\Delta L_2 - \Delta L_1)} \quad (10)$$

This term does not appear in the Y-branch MZI because the splitter is symmetric.

3. Modelling and simulation

3.1. Waveguide geometry and mode profile

The waveguide is a silicon strip 500 nm wide and 220 nm thick (fixed by the process) on a buried oxide (BOX) with SiO_2 cladding. It is designed for quasi-TE polarization at 1550 nm, compatible with the TE grating couplers of the library.

The mode was computed with the FDE solver of Lumerical MODE 2026 R1, using a 10 nm mesh and a $3 \times 2.2 \mu\text{m}$ simulation window. The material indices are described by Sellmeier formulas: Si from Li and SiO_2 from Malitson.

Fig. 1 shows the intensity profile of the TE_0 mode at 1550 nm: the TE polarization fraction is 98.4 % and 66 % of $|E|^2$ is confined in the core.

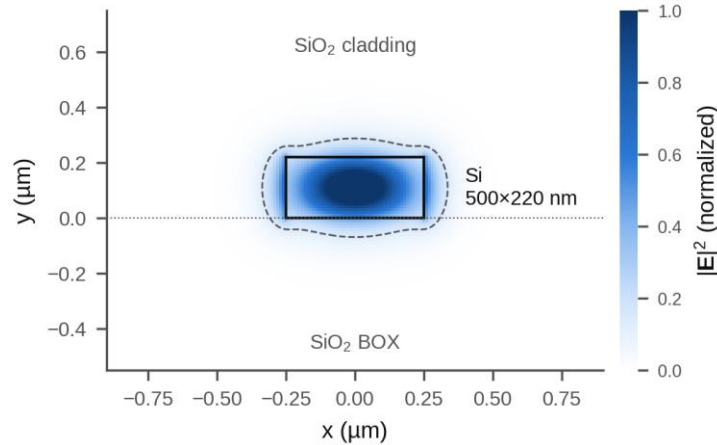


Fig. 1. Intensity profile $|E|^2$ of the TE_0 mode of the $500 \times 220 \text{ nm}$ strip waveguide at 1550 nm, computed with Lumerical MODE (FDE).

3.2. Effective and group indices

The wavelength sweep from 1500 to 1600 nm gives Fig. 2. The effective index decreases almost linearly from 2.501 to 2.389, while the group index changes little: 4.174 at 1500 nm, 4.182 at 1550 nm and 4.190 at 1600 nm. This value, $n_g \approx 4.18$, is the one we expect to extract from the devices.

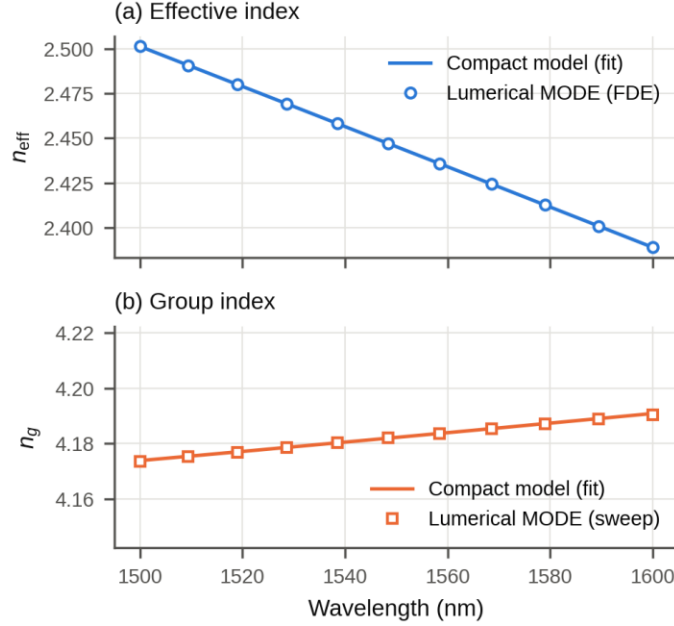


Fig. 2. (a) Effective index and (b) group index of the TE₀ mode versus wavelength (Lumerical MODE).

3.3. Waveguide compact model

For circuit simulation, $n_{\text{eff}}(\lambda)$ is represented by a second-order polynomial around $\lambda_0 = 1.55 \mu\text{m}$, fitted to the MODE results:

$$n_{\text{eff}}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2 \quad (11)$$

with $n_1 = 2.445$, $n_2 = -1.121 \mu\text{m}^{-1}$ and $n_3 = -0.055 \mu\text{m}^{-2}$ (λ in μm). Substituting (11) into (2) gives the group index of the model:

$$n_g(\lambda) = n_{\text{eff}}(\lambda) - \lambda[n_2 + 2n_3(\lambda - \lambda_0)] \quad (12)$$

At λ_0 , $n_g = n_1 - n_2\lambda_0 \approx 4.18$, in agreement with the MODE value, and the waveguide dispersion is $D = -(\lambda/c) d^2n_{\text{eff}}/d\lambda^2 = 5.69 \times 10^{-4} \text{ s/m}^2$.

3.4. Proposed circuits

Fig. 3 summarizes the three types of structure, which are simulated in Lumerical INTERCONNECT with the SiEPIC EBeam library. All of them use TE grating couplers to couple light in and out of the chip, 500 nm strip waveguides and 5 μm radius bends, whose losses are negligible for this waveguide [1]. In the simulation, the grating couplers are the ebeam_gc_te1550 library model, and in the layout, the GC_TE_1550_8degOxide_BB coupler of the process; their response is removed with the calibration structure, so the difference does not affect the extraction of n_g .

- *MZI (a)*: ebeam_y_1550 Y-branches. This is the default design of the course: robust and with high extinction.
- *Michelson (b)*: ebeam_bdc_te1550 BDC [3] and Y-branch loop mirrors. The BDC is needed because the reflected light must exit through a port different from the input; its nearly flat wavelength response keeps the splitting close to 50/50 across the band.
- *Calibration (c)*: two grating couplers connected by a straight waveguide. It measures the grating-coupler response, which is then subtracted from the interferometer spectra.

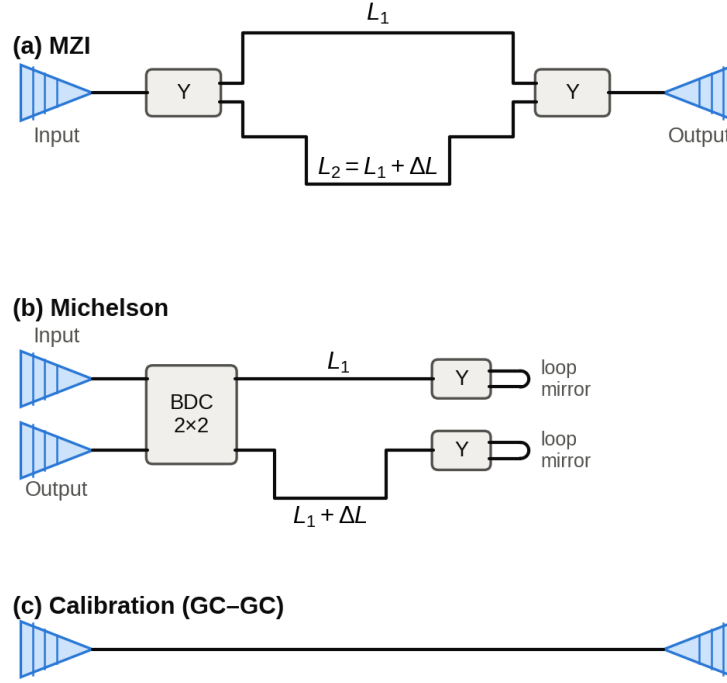


Fig. 3. Schematics of the proposed circuits: (a) MZI with Y-branches, (b) Michelson interferometer with a 2×2 BDC and loop mirrors, (c) grating-coupler (GC) calibration structure.

3.5. Parameter variations

Table 1 lists the nine devices, with the theoretical FSR from (7) and (8) using $n_g = 4.182$ and the FSR obtained in INTERCONNECT (Section 3.6). The ΔL values were chosen with two criteria:

- the largest FSR (11.5 nm, MZI1) leaves at least four fringes within the 50 nm measurement band;
- the smallest FSR (2.3 nm, MZI5) contains about 2300 points per fringe with the 1 pm minimum laser step, enough to locate the minima accurately.

The five MZIs allow the $\text{FSR} \propto 1/\Delta L$ trend to be studied, and MZI6 repeats MZI2 to estimate the variability between identical devices. MIC1 has almost the same FSR as MZI2 with half the ΔL , and MIC2 the same as MZI4, which tests the factor of 2 in (8); having two Michelson interferometers also allows (10) to be applied. The short-arm length L_1 will be the minimum allowed by the routing in the layout; only ΔL affects the FSR.

Table 1. Proposed devices and expected performance

Dev.	Type	Splitter	ΔL (μm)	Theoretical FSR (nm)	INTERCONNECT FSR (nm)	Fringes in 50 nm
CAL	GC–GC loop	—	—	—	—	—
MZI1	MZI	Y-branch	50	11.49	11.53	4.4
MZI2	MZI	Y-branch	100	5.74	5.74	8.7
MZI3	MZI	Y-branch	150	3.83	3.82	13.1
MZI4	MZI	Y-branch	200	2.87	2.87	17.4
MZI5	MZI	Y-branch	250	2.30	2.29	21.8
MZI6*	MZI	Y-branch	100	5.74	5.74	8.7
MIC1	Michelson	BDC + loop	50	5.74	5.88	8.7
MIC2	Michelson	BDC + loop	100	2.87	2.89	17.4

*Replica of MZI2 to assess fabrication variability. Theoretical FSR with $n_g = 4.182$ at 1550 nm; INTERCONNECT FSR with the EBeam library waveguide, at the minimum closest to 1550 nm. All devices: 500×220 nm strip waveguide, TE.

3.6. Circuit simulation in INTERCONNECT

Each circuit was simulated with its own optical network analyzer (ONA) from 1500 to 1600 nm (20001 points, ≈ 5 pm step), including the grating couplers, as in the real measurement. Two waveguide models were used: (i) the ebeam_wg_integral_1550 library waveguide, which is the one the layout will use ($n_{\text{eff}} = 2.446$, $n_g = 4.191$, $D = 3.54 \times 10^{-4}$ s/m², 7 dB/cm), and (ii) a “Straight Waveguide” element with the MODE results ($n_{\text{eff}} = 2.445$, $n_g = 4.182$, $D = 5.69 \times 10^{-4}$ s/m², 3 dB/cm assumed).

Fig. 4 shows the spectra obtained with the library waveguide. The calibration structure (CAL) shows a loss of 4.7 dB for the two grating couplers, with the peak at 1554 nm and a -3 dB bandwidth of 44 nm, confirming the usable measurement band of about 50 nm. The MZI extinction ratio exceeds 55 dB. For the same ΔL , the Michelson interferometer (MIC1) shows twice as many fringes as the MZI (MZI1). Table 1 compares the FSR values obtained with theory.

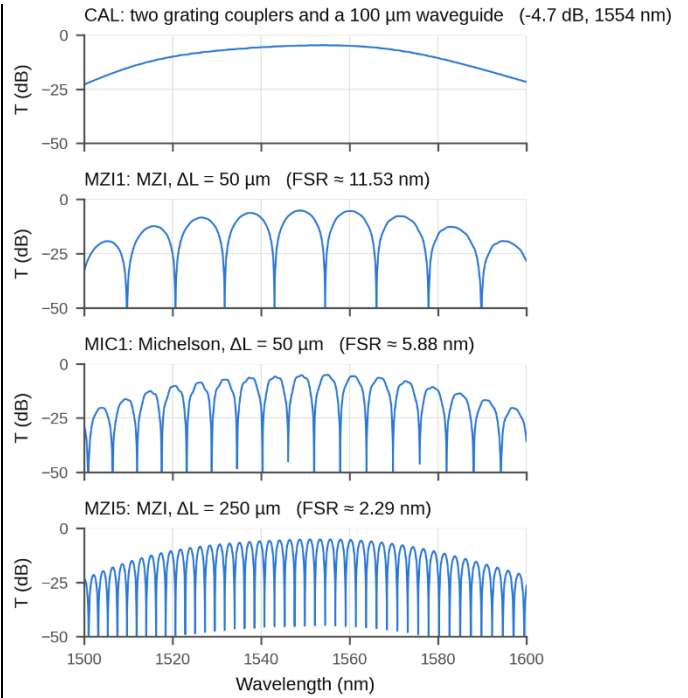


Fig. 4. Transmission spectra simulated in Lumerical INTERCONNECT for the calibration structure, MZI1, MIC1 and MZI5.

3.7. Test of the extraction equation

To test (7), (8) and (10) and the analysis procedure, the INTERCONNECT spectra were treated as if they were measurements: the minima were located with parabolic interpolation, the FSR was computed between consecutive minima, and the corresponding equation was applied at the central wavelength of each pair. Table 2 summarizes the values near 1550 nm.

Table 2. Group index extracted from the INTERCONNECT spectra (near 1550 nm)

Device	ΔL (μm)	FSR, EBeam waveguide (nm)	n_g , EBeam waveguide	FSR, MODE waveguide (nm)	n_g , MODE waveguide
MZI1	50	11.531	4.191	11.553	4.183
MZI2	100	5.744	4.191	5.755	4.182
MZI3	150	3.825	4.191	3.832	4.182
MZI4	200	2.867	4.191	2.873	4.182
MZI5	250	2.293	4.191	2.297	4.182
MIC1, eq. (8)	50	5.877	4.098	5.889	4.090
MIC2, eq. (8)	100	2.894	4.152	2.900	4.143
MIC1 + MIC2, eq. (10)	50 / 100	—	4.206	—	4.197
Reference (waveguide model)	—	—	4.191	—	4.182

The reference row is the n_g at 1550 nm of each waveguide model used in INTERCONNECT.

For the MZIs, the extracted n_g matches that of the waveguide model in both cases (4.191 and 4.182), so the approximation in (7) introduces no significant error. For the Michelson interferometers, (8) gives values 2 % (MIC1) and 1 % (MIC2) lower, because of the BDC term δ (Section 2.5), which turns out to be $\delta \approx -10.8 \mu\text{m}$. Combining MIC1 and MIC2 with (10) reduces the error to 0.4 %. Therefore, the MZIs will be the main source of n_g , and the Michelson interferometers will be analyzed in pairs.

3.8. Procedure for the experimental data

With the measured spectra, n_g will be obtained in the following steps:

- 1) normalize each spectrum by subtracting (in dB) that of the calibration structure, to remove the grating-coupler envelope;
- 2) locate the interference minima, which are narrower than the maxima, with a local fit, and compute the FSR between adjacent minima;
- 3) compute $n_g(\lambda)$ with (7) for the MZIs;
- 4) as a more robust method, fit the transfer function (4) to each MZI spectrum with n_1 , n_2 and n_3 as free parameters, which gives $n_{\text{eff}}(\lambda)$ and $n_g(\lambda)$ at once;
- 5) plot $1/\text{FSR}$ versus ΔL for the five MZIs: the slope is n_g/λ^2 , and the linear fit averages out the error of each individual device;
- 6) for the Michelson interferometers, use (10) with MIC1 and MIC2 and compare with the MZIs;
- 7) compare MZI2 with MZI6 to estimate variability and compare the result with simulation; differences will be attributed to deviations of the fabricated width or thickness.

4. Fabrication

The draft layout (EBeam_JuPR65.oas), with the nine devices of Table 1, was designed with KLayout and SiEPIC-Tools and the SiEPIC EBeam PDK, and is submitted to the openEBL-2026-10 repository for peer review. In the draft, the ΔL values differ by at most 1 μm from the design values in Table 1.

To be completed in the final layout phase (October 17, 2026).

5. Experimental data

To be completed in the measurement phase.

6. Analysis

To be completed in the analysis phase.

7. Conclusion

To be completed in the final report (December 8, 2026).

References

1. L. Chrostowski and M. Hochberg, *Silicon Photonics Design: From Devices to Systems* (Cambridge University Press, 2015).
2. L. Chrostowski, "Compact photonic models" and "Photonic circuits built using compact models," lecture notes, *Silicon Photonics Design, Fabrication and Data Analysis* (UBCx Phot1x, edX, 2026).
3. Z. Lu, H. Yun, Y. Wang, Z. Chen, F. Zhang, N. A. F. Jaeger, and L. Chrostowski, "Broadband silicon photonic directional coupler using asymmetric-waveguide based phase control," *Opt. Express* **23**(3), 3795–3808 (2015).

To be completed in the final report (December 8, 2026).